

ROAD HEALTH INDEX (RHI) & REMAINING SERVICE LIFE SYSTEM

An AI-Powered Dual-Track Machine Learning Platform for Proactive Infrastructure Asset Management

Academic Institution: Acharya Institute of Technology	Department: Artificial Intelligence & Machine Learning (AI & ML)
Project Group: Group 18	Project Guide: Mr. Mohammed Tahir Mirji (Assistant Professor)
Primary Data Source: FHWA Long-Term Pavement Performance (LTPP)	Core Models: Supervised XGBoost Regressor + Unsupervised K-Means

1. Executive Summary: The Infrastructure Crisis & The AI Solution

Highways and municipal roadways deteriorate continuously under two relentless forces: **mechanical traffic stress** (repetitive heavy truck axle passes) and **environmental climate stress** (freeze-thaw thermal expansion, severe cold cycles). Traditional road management relies on manual visual inspections, which are dangerous, subjective, and reactive. This project delivers an end-to-end machine learning platform that evaluates non-destructive testing (NDT) sensor scans to calculate an objective **0–100 Road Health Index (RHI)** and forecast **Remaining Service Life (RSL)**.

Traditional Road Inspection (Manual & Flawed)	The AI-Powered RHI Solution (Automated & Proactive)
Dangerous & Slow: Human inspectors walk or drive along high-speed corridors.	100% Automated & Safe: Evaluates road segments instantly using non-destructive sensor scans.
Subjective & Inconsistent: Different inspectors assign conflicting condition scores to the same road.	Objective & Standardized: Computes a unified 0–100 index grounded in physics and ML.
Reactive (Pothole-Driven): Repairs occur only after visible collapse, costing up to 5× more.	Proactive (Prescriptive): Forecasts deterioration 10 years ahead to schedule low-cost preventive overlays.

2. Essential Engineering Concepts Explained Simply

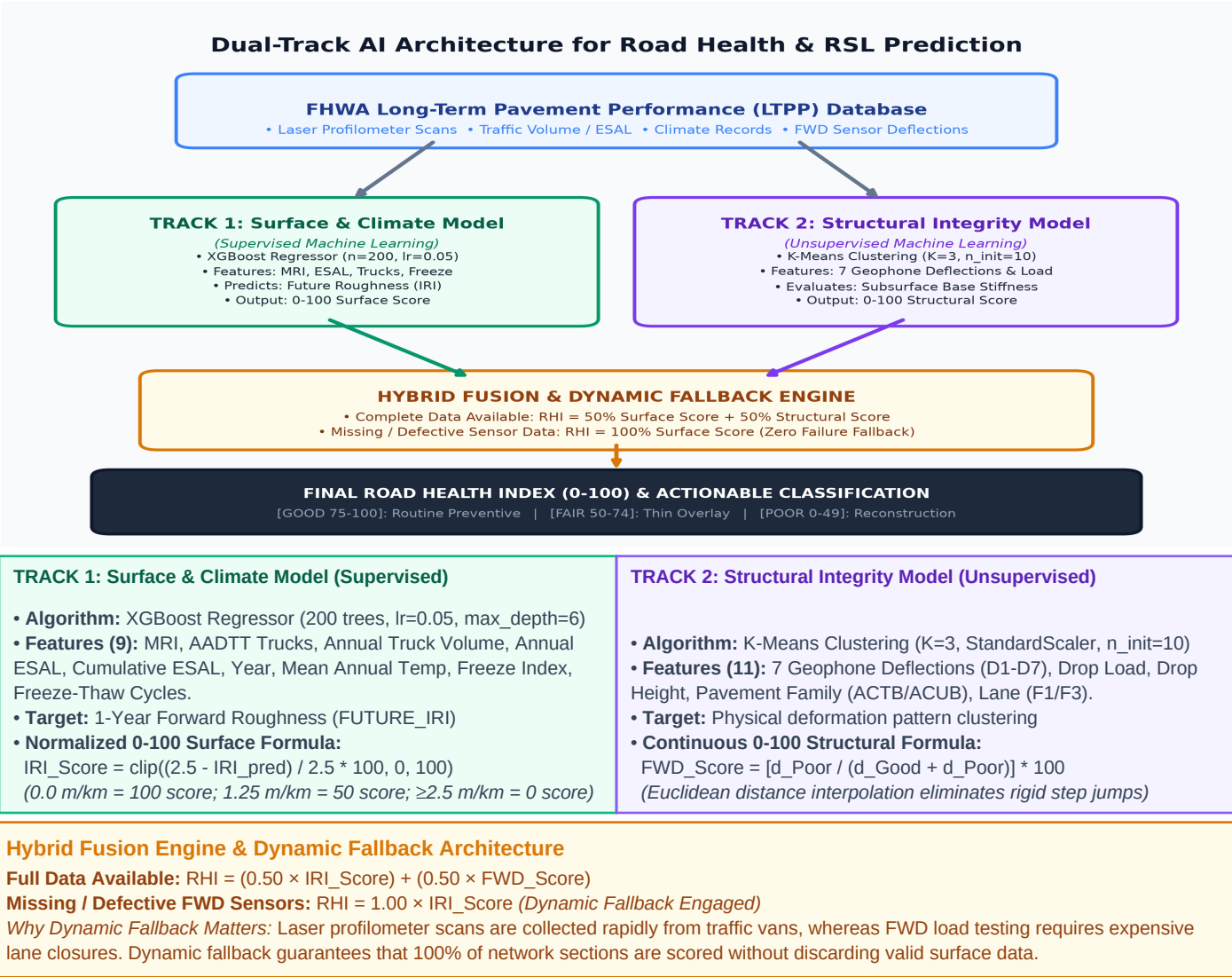
Term	Everyday Meaning	Civil & Machine Learning Significance
IRI / MRI (Roughness)	How bumpy or rough the ride feels inside a moving vehicle.	Measured in m/km by laser profilometer vans. IRI > 2.5 m/km is classified by FHWA as critically failed.
FWD (Falling Weight)	A trailer that drops a heavy load on the road to measure underground flex.	7 geophone sensors record instantaneous peak deflections (in microns, µm) to evaluate base stiffness.
Deflection Basin	The bowl-shaped dent created by the 7 sensors when struck by the load.	Shallow bowl = Rock-solid foundation. Deep bowl = Soft, fatigued, or cracked subgrade base.
ESAL (Axle Load)	Converts all mixed traffic into standard 18,000-lb truck passes.	1 loaded semi-truck causes structural damage equivalent to ~5,000 passenger cars.
Freeze-Thaw	Water in cracks freezes (+9% expansion) and thaws, shattering asphalt.	Annual thermal cycles rapidly accelerate surface micro-cracking and water penetration.
RSL (Remaining Life)	Number of years before the road degrades past the failure threshold.	Allows transportation agencies to allocate funding years before catastrophic collapse.

Economic Takeaway: Proactive Maintenance vs Reactive Reconstruction

Applying a thin preventive asphalt overlay when a road is in **Fair condition (RHI 50-74)** costs ~\$15-25/sq.yard and extends life by 10+ years. Waiting until **Poor condition (RHI < 50)** forces full-depth structural reconstruction costing ~\$120-150/sq.yard (**5× to 8× higher**).

3. Dual-Track Machine Learning Architecture & Scoring Logic

Road health comprises two distinct layers: **(1) Surface Ride Quality** (measured by laser profilometers) and **(2) Subsurface Structural Health** (measured by falling weight deflectometers). Our architecture trains two specialized machine learning models and combines them through a hybrid fusion engine:



4. Pavement Condition Classification & Prescriptive Actions

RHI Score	Rating	Physical Status	Recommended Engineering Action	Cost
75.0 – 100.0	GOOD	Smooth ride, minimal roughness, shallow deflection basin, high base stiffness.	Routine inspection, crack sealing, localized fog seal, preventive maintenance.	\$ (Low)
50.0 – 74.9	FAIR	Moderate surface wear, micro-rutting, minor structural deflection fatigue.	Schedule thin asphalt overlay (1.5-2 in), micro-surfacing, or base patching.	\$\$ (Med)
0.0 – 49.9	POOR	Severe roughness (IRI ≥ 2.0 m/km), deep deflection basin, base layer failure.	Immediate structural rehabilitation, deep mill-and-fill, or full reconstruction.	\$\$\$\$\$ (5x)

5. Visualizing Non-Destructive Testing & 10-Year Forecasts

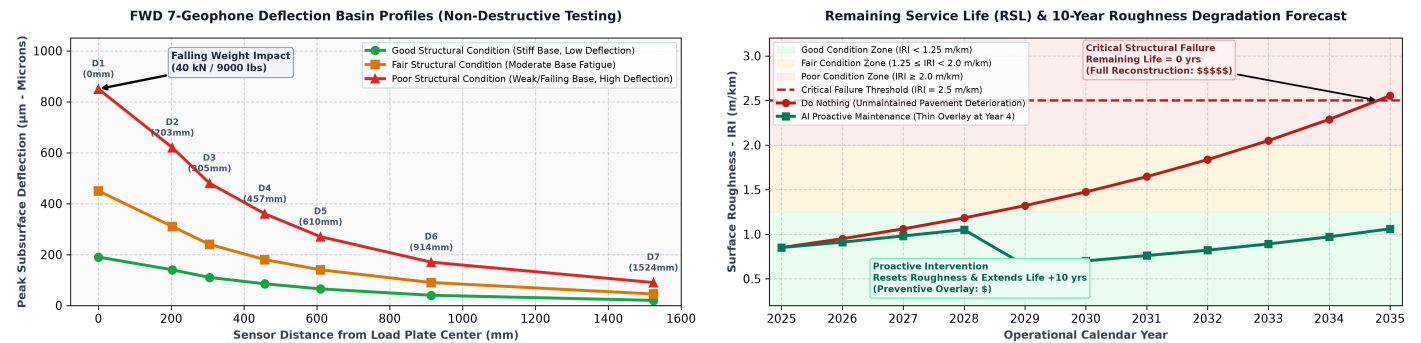


Figure A: FWD 7-Geophone Deflection Basin Profiles
Sensor D1 is under the load plate; D7 is 1.5m away. A stiff, healthy road deflects minimally (<200 µm), while a broken base deflects deeply (>800 µm).

Figure B: Remaining Service Life & Proactive Intervention
Unmaintained roads rapidly cross the 2.5 m/km failure threshold. A timely thin overlay at Year 4 resets roughness and extends road life by 10+ years at minimal cost.

6. Software Ecosystem & Web Control Center

The project provides a comprehensive software delivery stack, from interactive terminal tools to a modern web application:

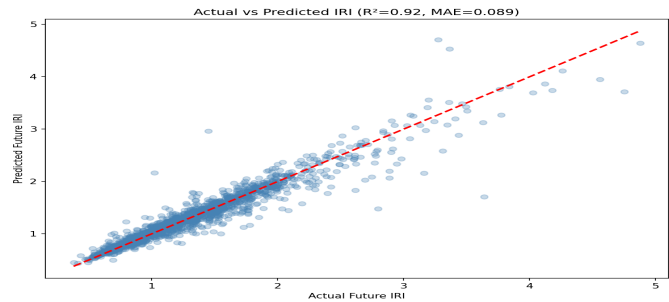
Component / Module	Technology	Key Features & Capabilities
FastAPI Backend Service (Dashboard/main.py)	Python 3, FastAPI, Uvicorn, Pydantic	- High-speed REST API endpoints (/api/predict, /api/sections, /api/batch). - LRU-cached model artifact loader for sub-millisecond inference. - Tree SHAP explainability engine extracting top feature contributions.
Web Control Center (Dashboard/static/)	HTML5, Vanilla JS, CSS3, Chart.js	- Dynamic SVG animated radial gauge for real-time RHI visualization. - Interactive section search with historical defaults autofill. 10-Year forecast chart with red failure threshold overlay. - Deflection basin geometry visualizer with ±10% confidence bands.
Batch Processor & Reporting (api/batch, /api/report.csv)	Pandas, Python-Multipart, jsPDF	- Upload CSV/Excel files with up to 50 road segments for bulk scoring. - Instant client-side PDF Executive Assessment Card generator. - Downloadable master CSV inspection logs.
Interactive CLI Predictor (src/rhi_predictor.py)	Python Terminal CLI	- Standalone interactive terminal tool prompting for custom road parameters. - Instant ASCII condition scorecard output without needing a web browser.

7. Explainable AI (XAI) & 10-Year Projections

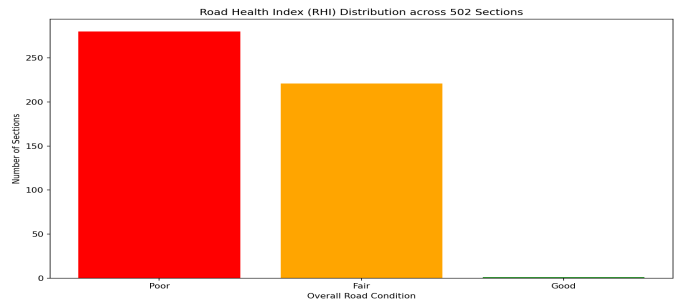
To earn civil engineers' trust, the system does not operate as an opaque black box. When a prediction is made, the **SHAP Tree Explainer** calculates the percentage impact of every input variable. For example:

Feature	Impact %	Direction of Impact	Engineering Interpretation
Cumulative ESAL	38.4%	Increases Roughness Risk	Historical heavy freight accumulation is the primary driver of structural wear.
Current MRI	29.1%	Increases Roughness Risk	Existing surface roughness propagates into faster micro-crack formation.
Freeze-Thaw Cycles	18.2%	Increases Roughness Risk	Water expansion in sub-zero winter temperatures fractures asphalt binders.
Mean Ann Temp	14.3%	Reduces Roughness Risk	Moderate, stable annual temperatures minimize thermal contraction cracking.

8. Model Evaluation & Experimental Findings



Model 1: Actual vs Predicted Future Roughness (R²=0.92, MAE=0.089)



Network Distribution across 502 LTPP Road Sections

9. Complete Codebase & Repository Directory Catalog

Directory / File Path	Type & Purpose in System
Dashboard/main.py	FastAPI REST backend server, prediction endpoints, SHAP explainability, batch evaluator.
Dashboard/static/	Web UI frontend: index.html (layout), app.js (charts/logic), styles.css (design system).
data/	Raw LTPP datasets (Profilometer roughness, Traffic volume trends, ESAL loads, Climate, FWD drops).
models/	Serialized artifacts: iri_prediction_model.pkl, fwd_kmeans_model.pkl, scalers, and encoders.
notebooks/	Research notebooks: model1.ipynb (XGBoost), model2.ipynb (K-Means), RHI_Score.ipynb (Fusion pipeline).
src/train_model1.py & train_model2.py	Standalone CLI training scripts to train and serialize Model 1 and Model 2.
src/rhi_predictor.py	Interactive terminal CLI tool for on-demand single-section Road Health prediction.
outputs/rhi_scores.csv	Master scored dataset containing evaluated RHI scores and condition ratings for all 502 LTPP sections.

10. Quickstart Guide: How to Run the Platform

1. Install Dependencies	pip install -r requirements.txt	3. Run Terminal CLI	python src/rhi_predictor.py
2. Start Web Server	python -m uvicorn Dashboard.main:app --reload --port 8000	4. Re-Train Models	python src/train_model1.py

11. Frequently Asked Questions (FAQ)

Q: Why K-Means for structural health instead of supervised classification? A: Subsurface road layers do not have ground-truth human labels. K-Means discovers natural stiffness clusters purely from physical deflection basins, eliminating human label bias.	Q: What if FWD deflection data is missing? A: Dynamic Fallback automatically computes RHI = 1.0 × IRI_Score. The system never crashes and scores 100% of network segments.
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Project Attribution & Academic Affiliation: Developed by **Group 18**, Department of Artificial Intelligence & Machine Learning (AI & ML), **Acharya Institute of Technology**, guided by **Mr. Mohammed Tahir Mirji** (Assistant Professor). Data source: US Federal Highway Administration (FHWA) Long-Term Pavement Performance (LTPP) Program.